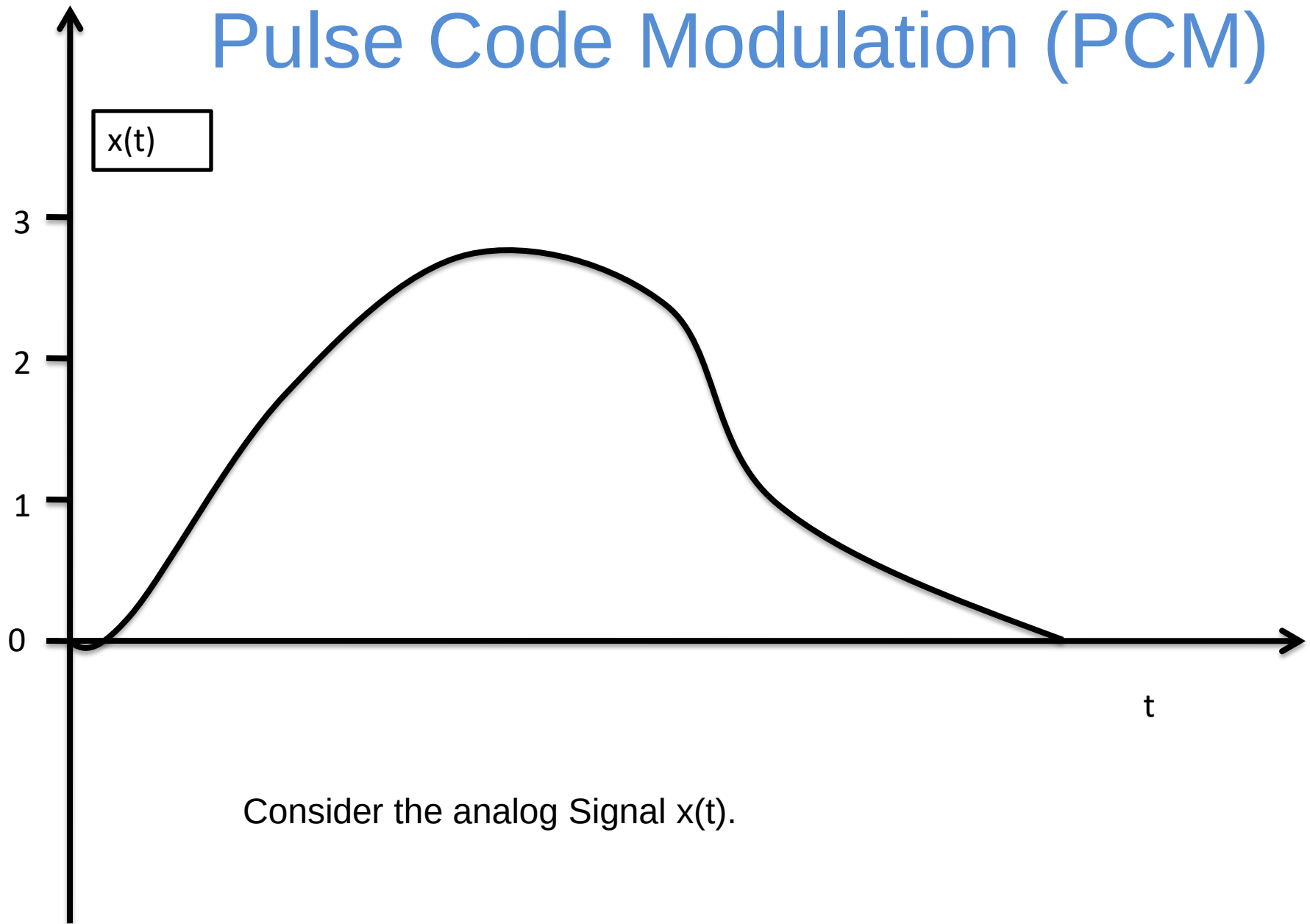
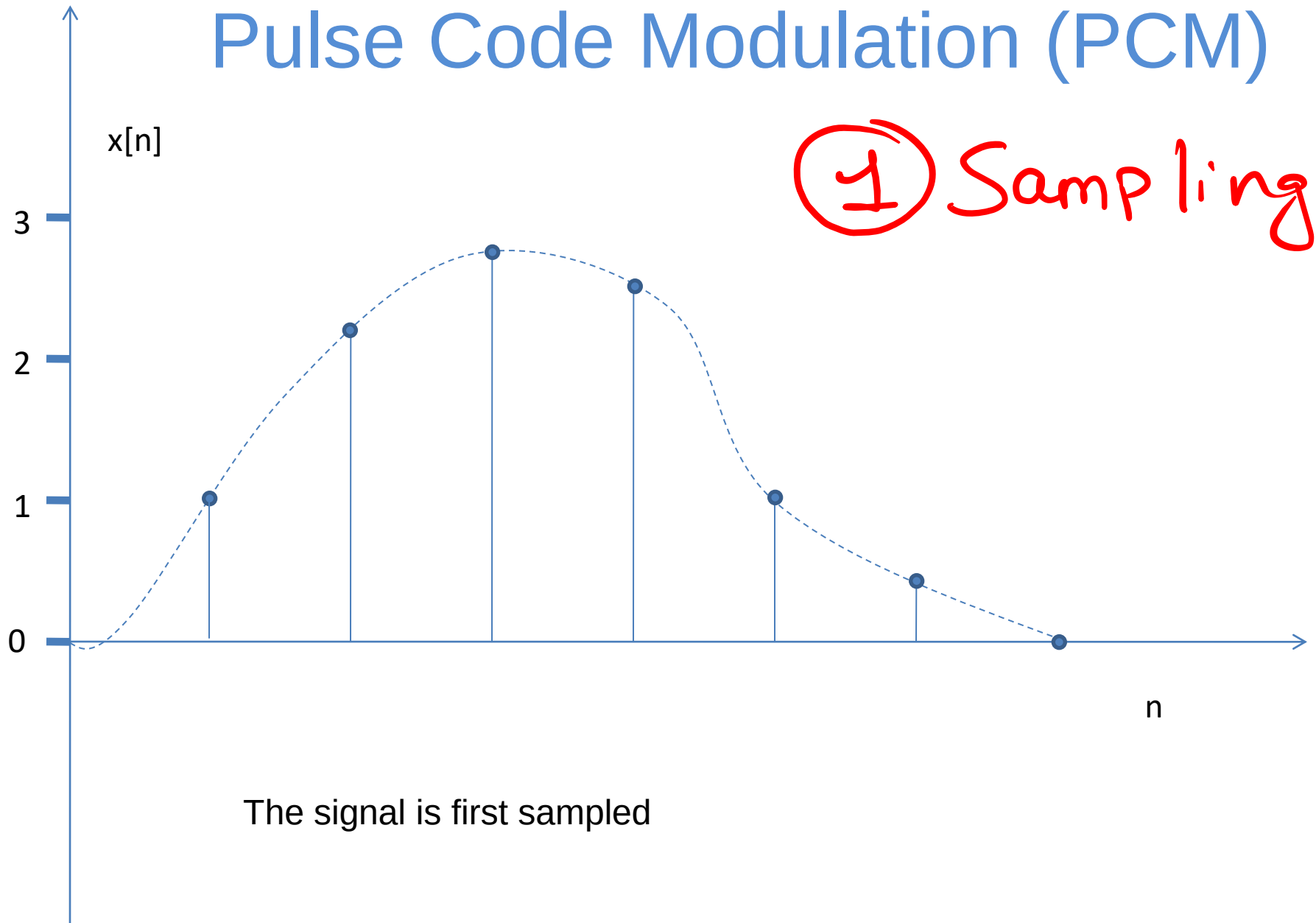


# Pulse Code Modulation (PCM)

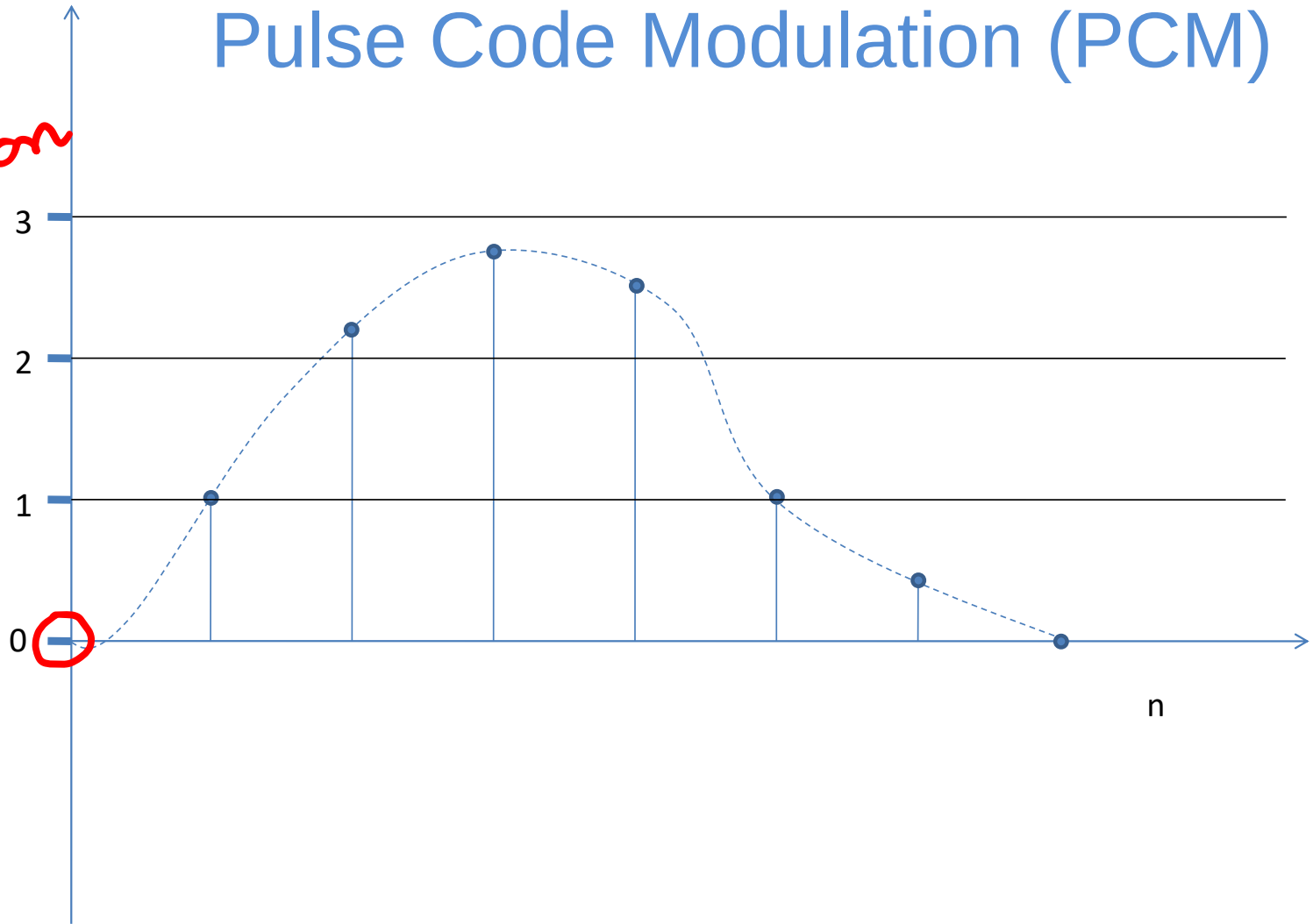


# Pulse Code Modulation (PCM)

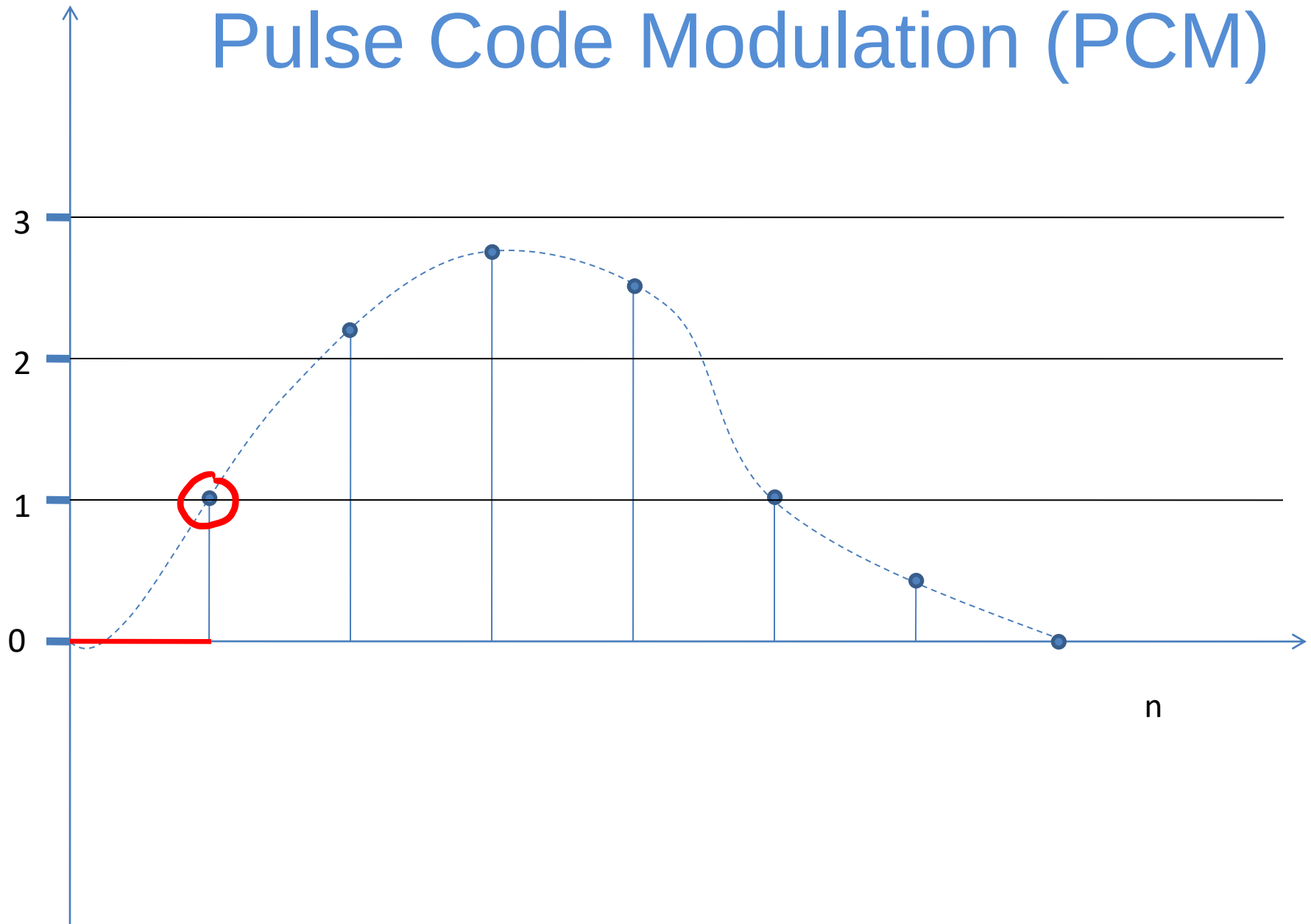


# Pulse Code Modulation (PCM)

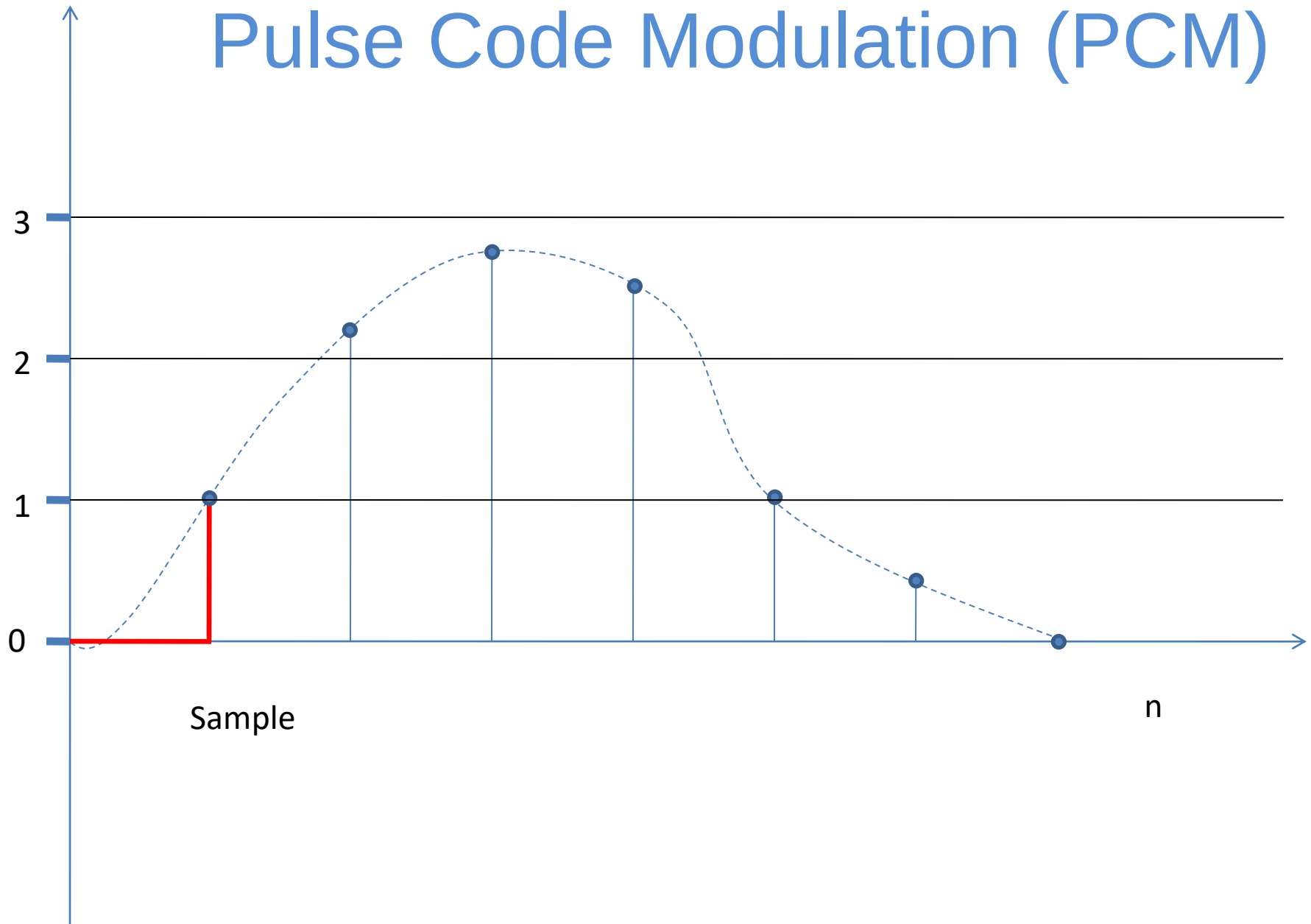
Quantization  
Levels



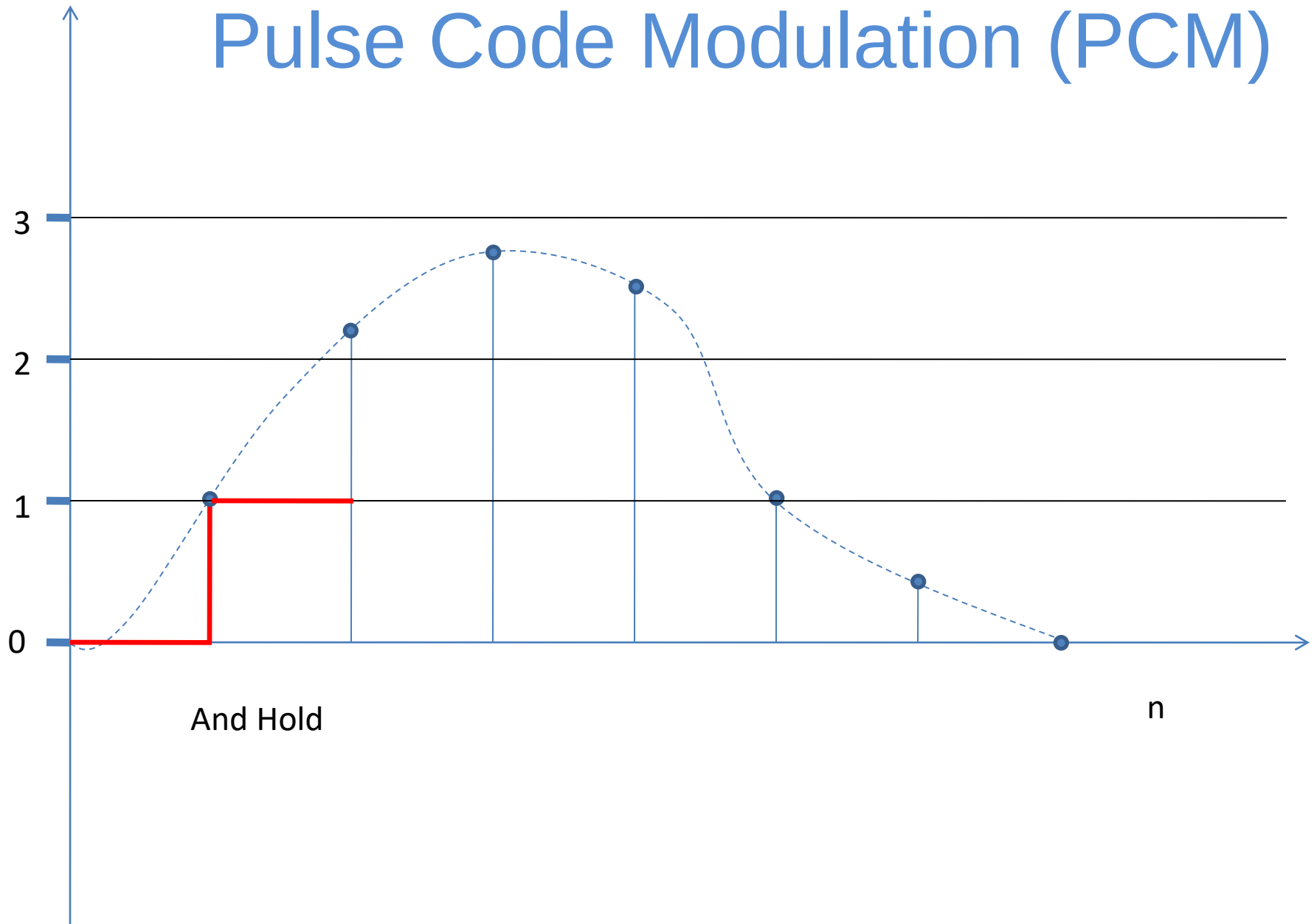
# Pulse Code Modulation (PCM)



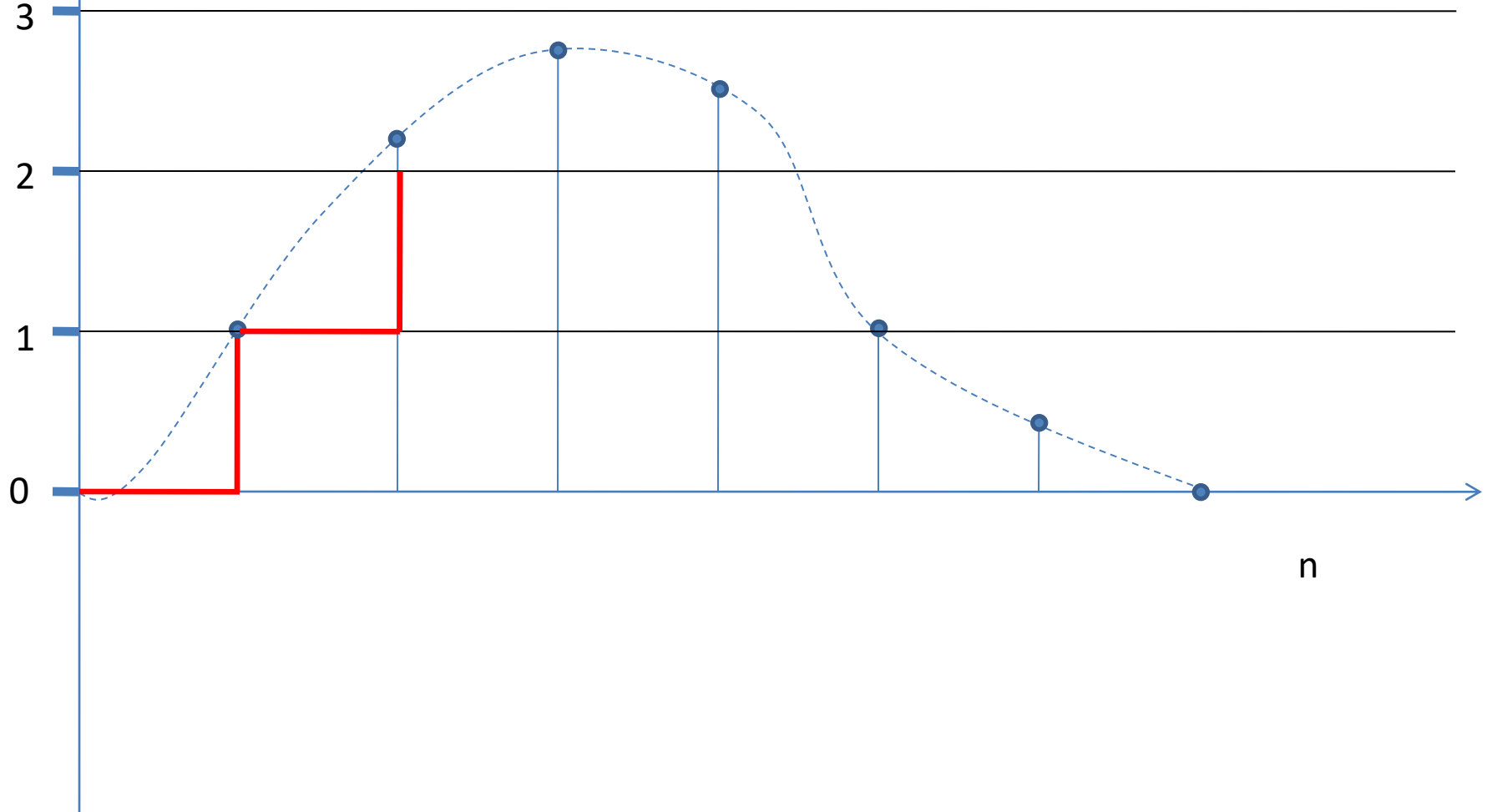
# Pulse Code Modulation (PCM)



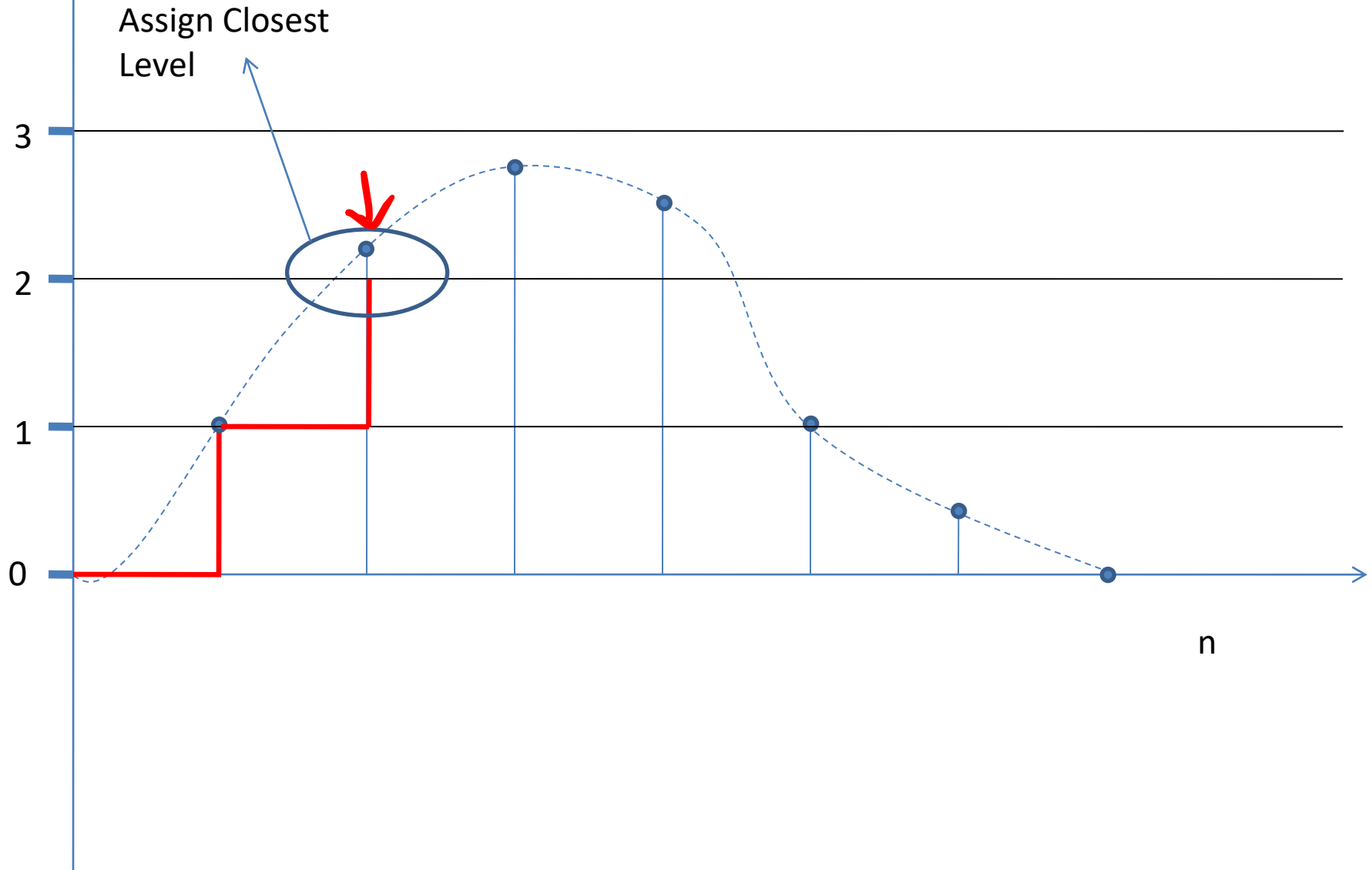
# Pulse Code Modulation (PCM)



# Pulse Code Modulation (PCM)

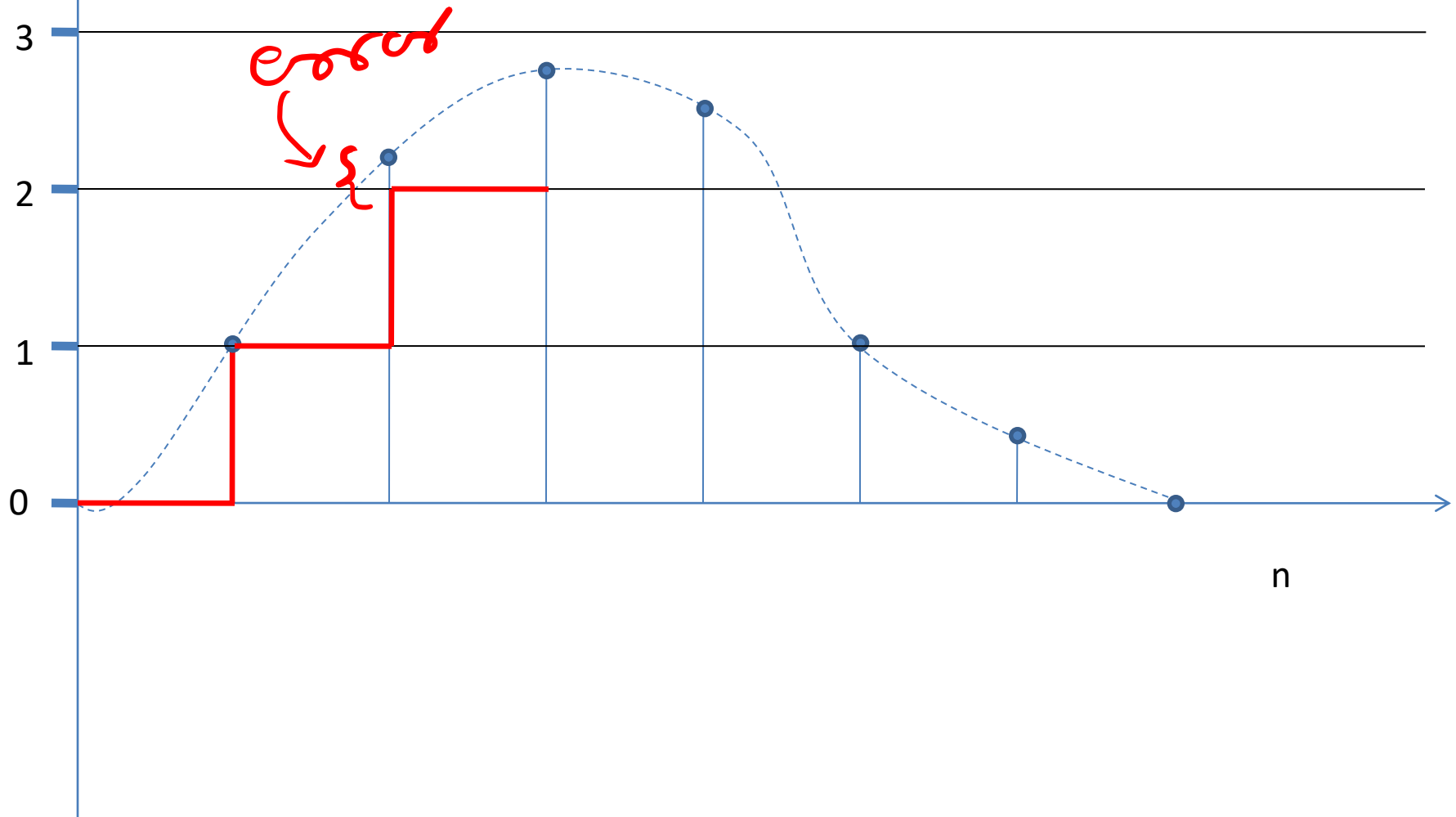


# Pulse Code Modulation (PCM)

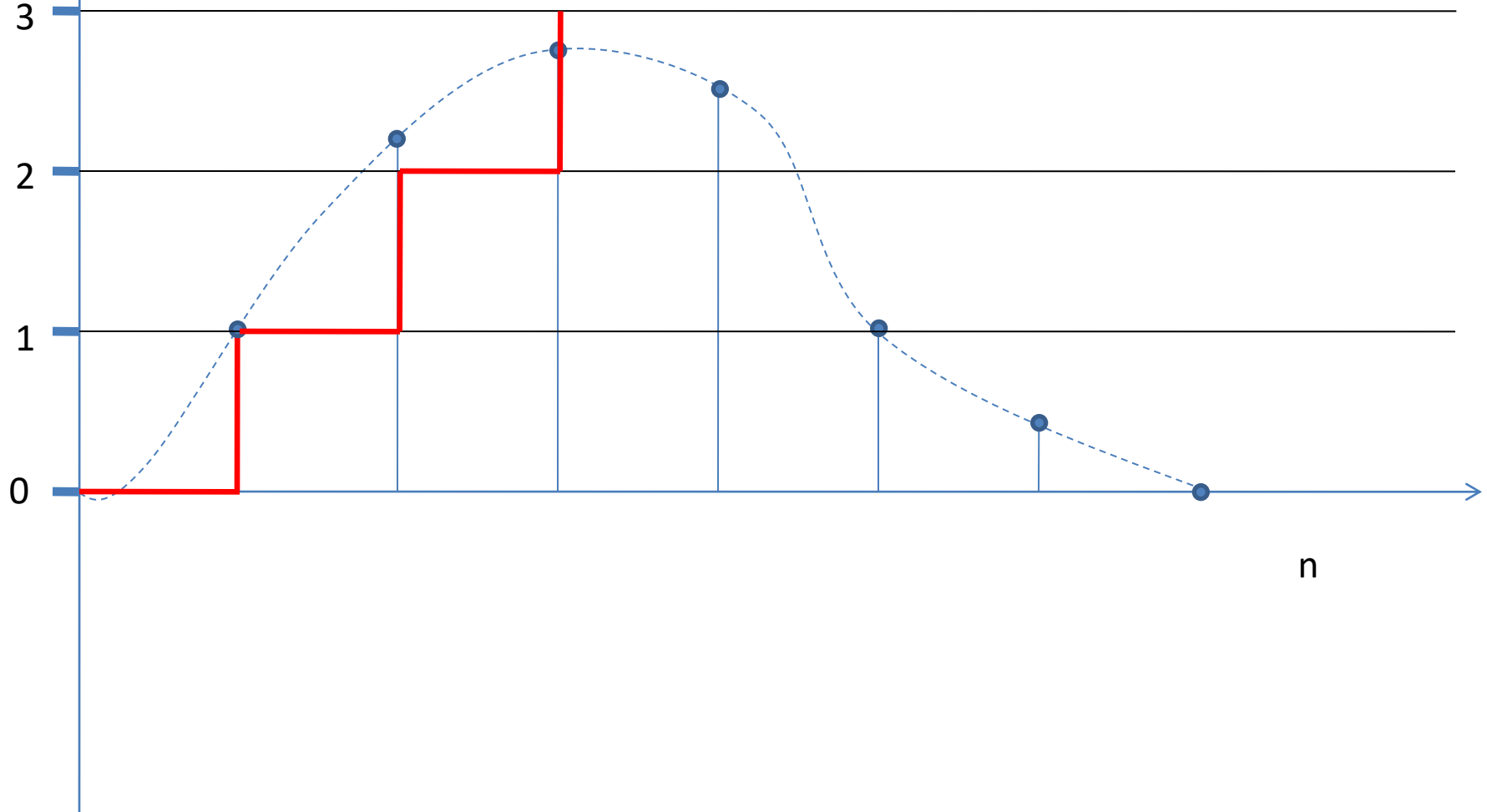




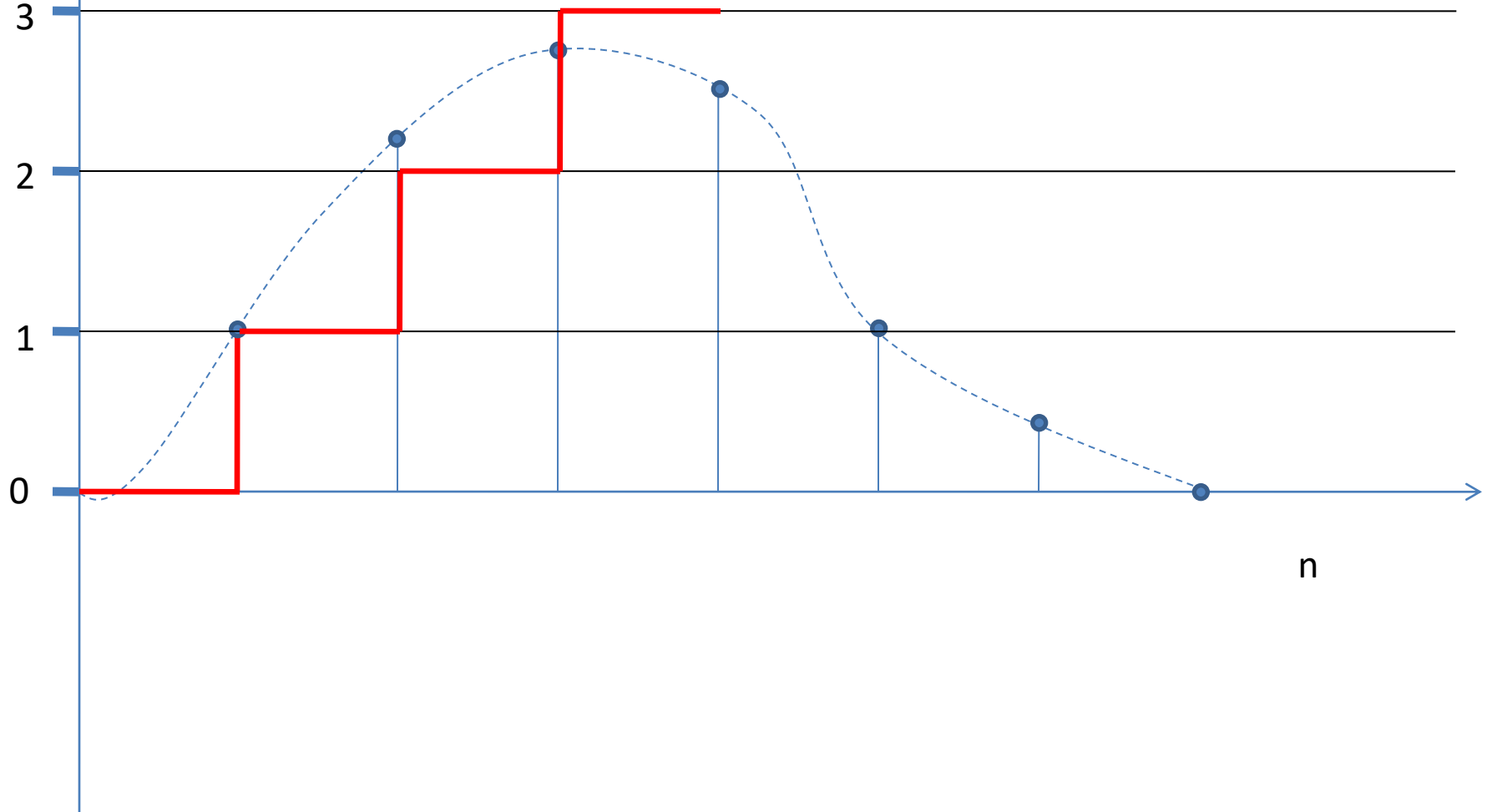
# Pulse Code Modulation (PCM)



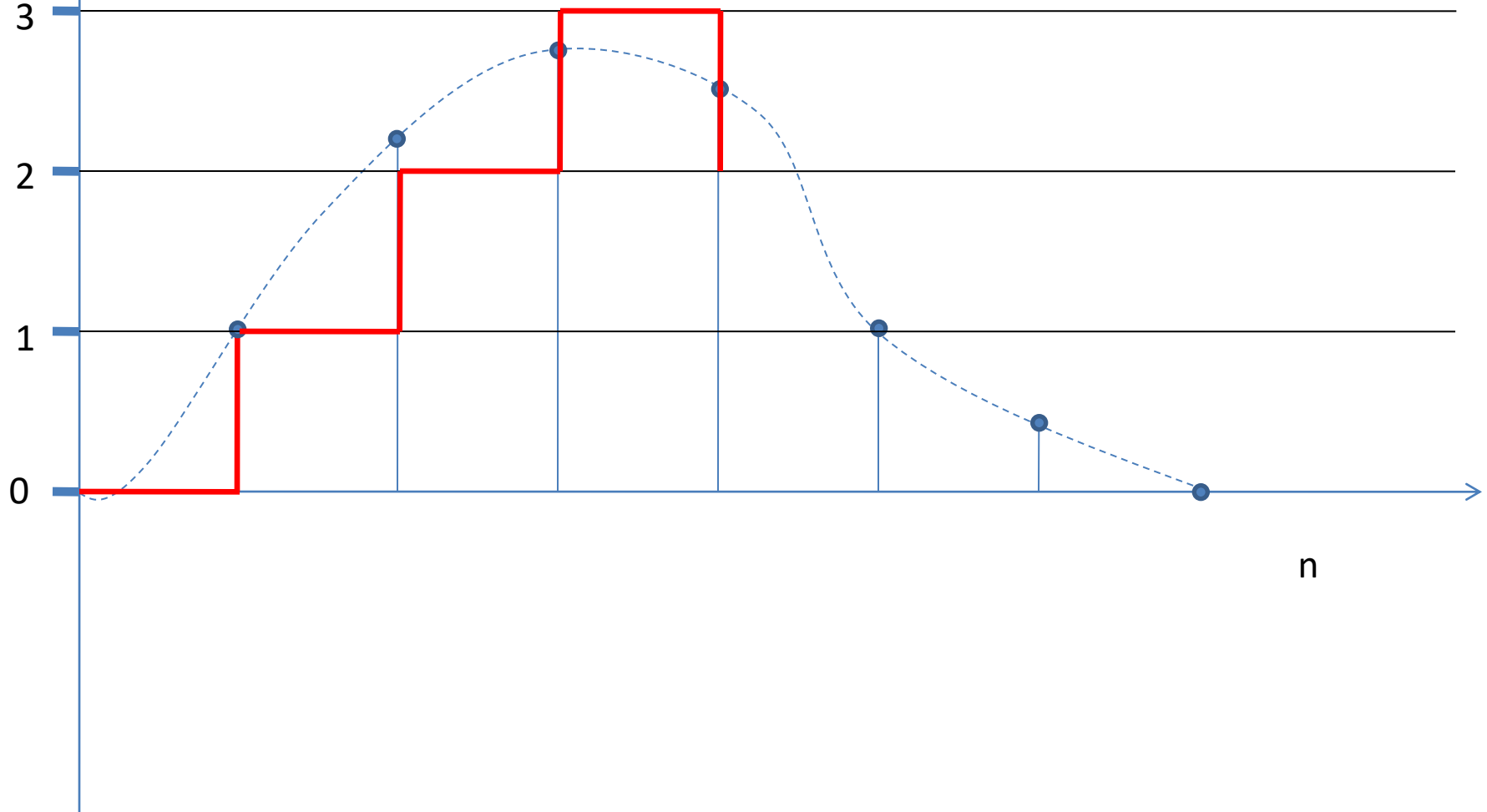
# Pulse Code Modulation (PCM)



# Pulse Code Modulation (PCM)

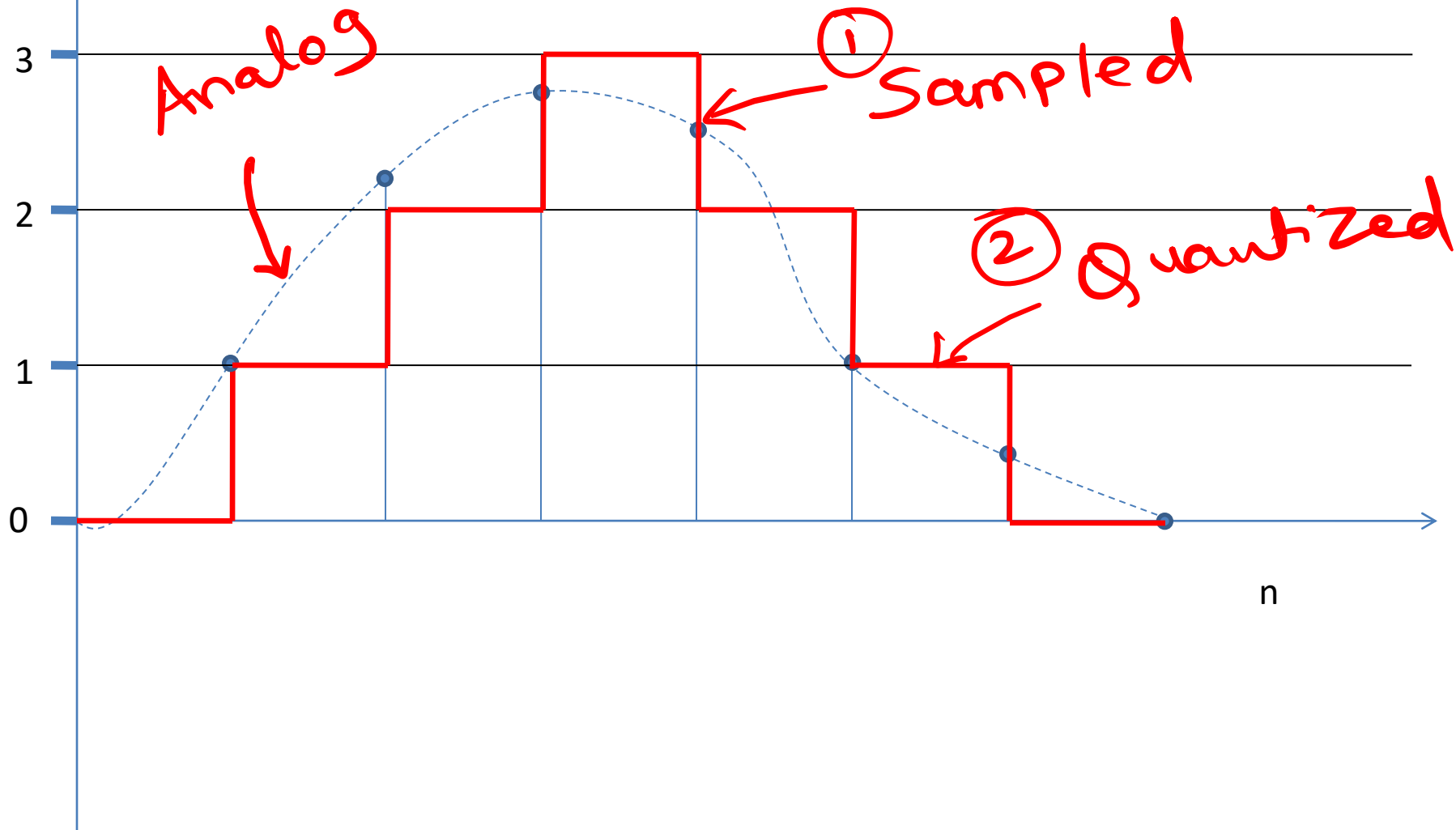


# Pulse Code Modulation (PCM)



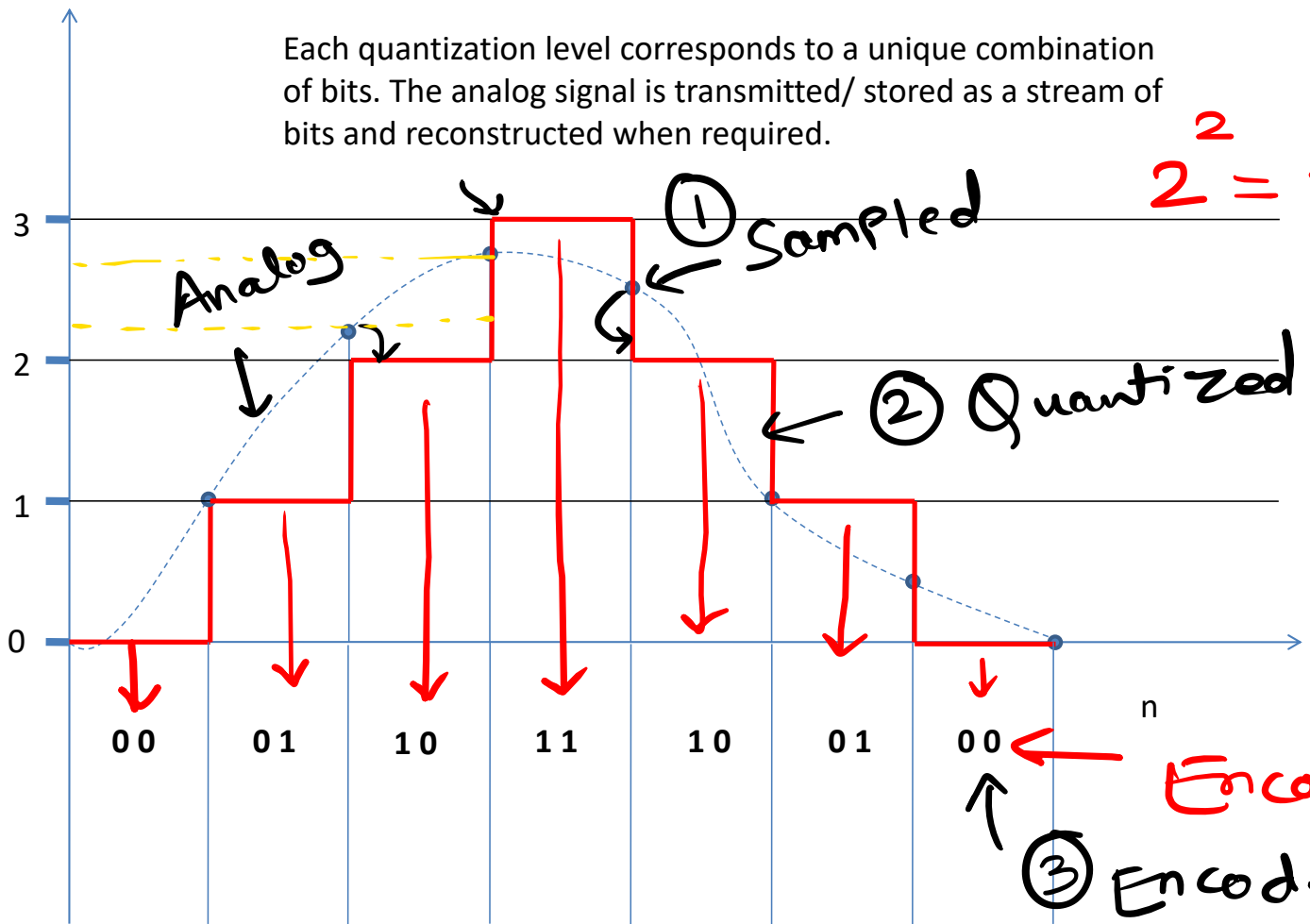


Each quantization level corresponds to a unique combination of bits. The analog signal is transmitted/ stored as a stream of bits and reconstructed when required.



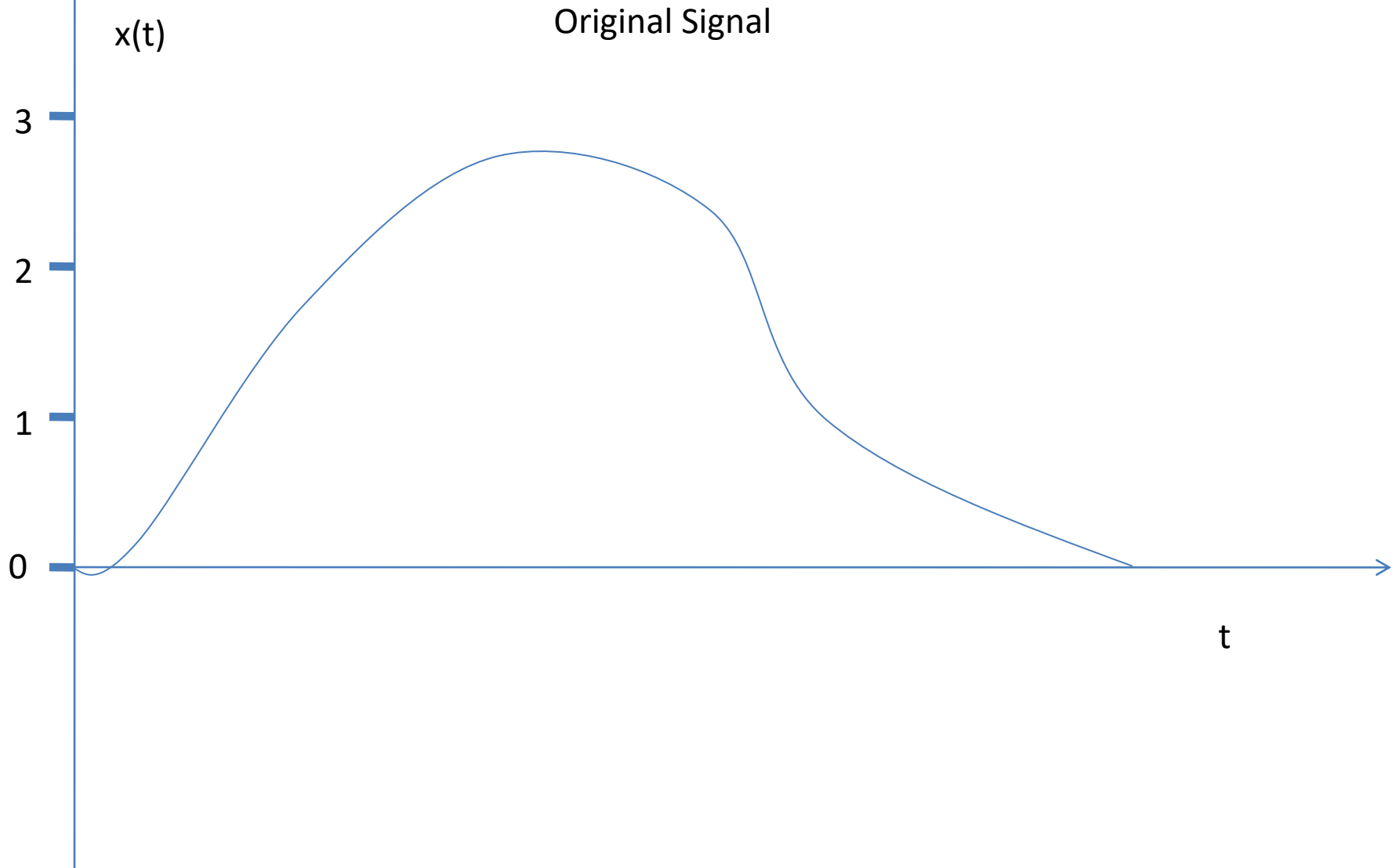
Each quantization level corresponds to a unique combination of bits. The analog signal is transmitted/ stored as a stream of bits and reconstructed when required.

Unique Binary code  
11  
10  
01  
00



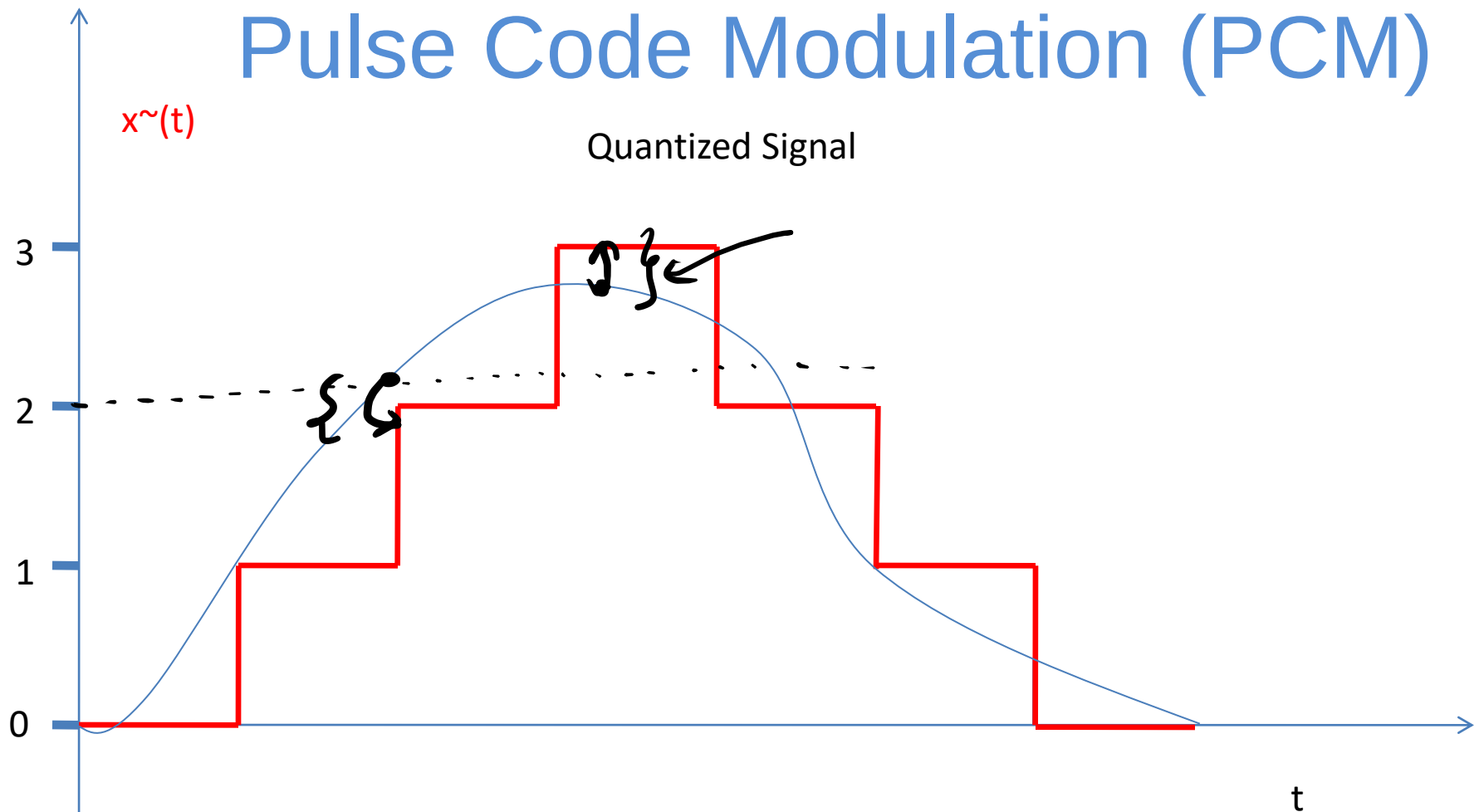
$$2^2 = 4$$

# Pulse Code Modulation (PCM)



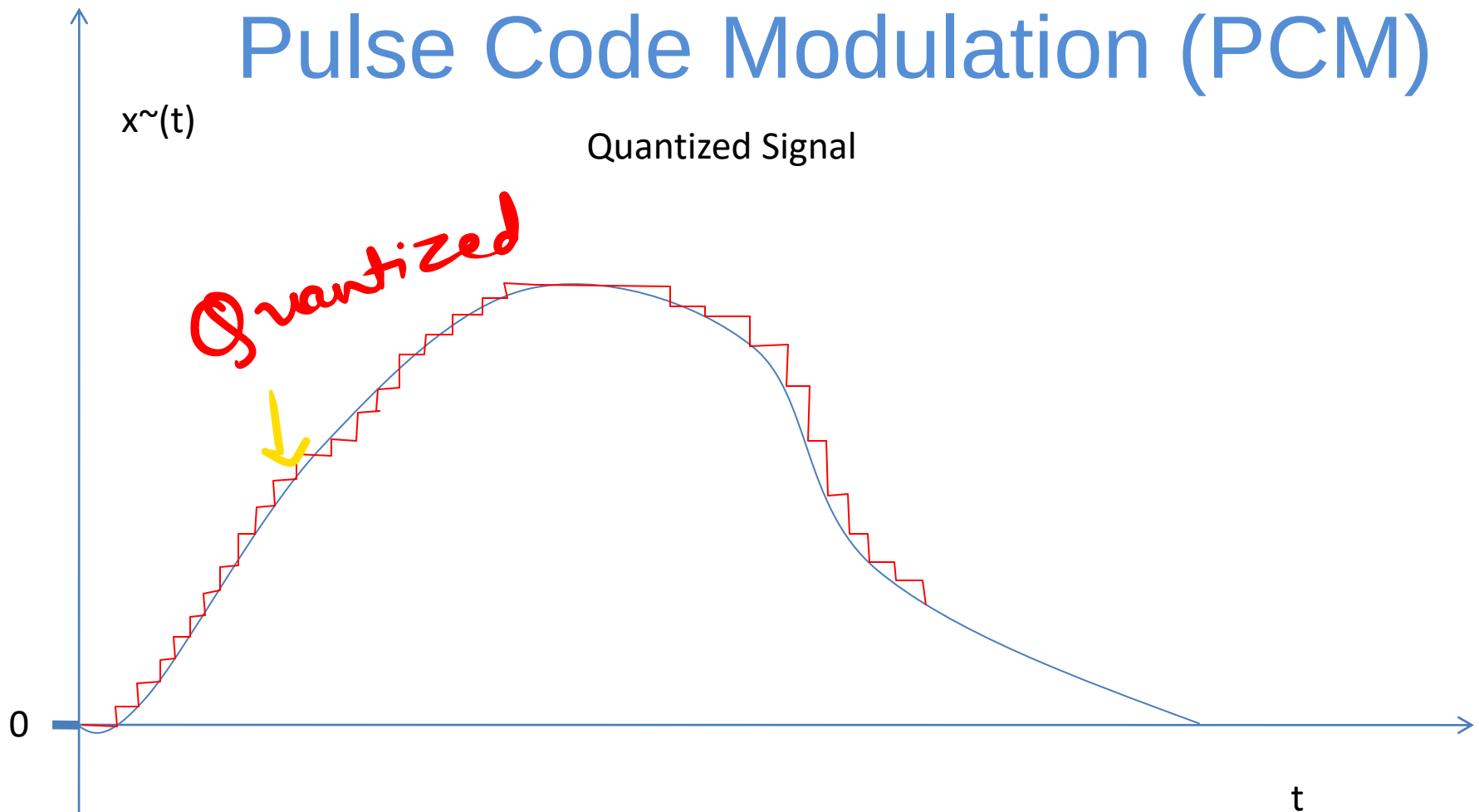


# Pulse Code Modulation (PCM)



It is quite apparent that the quantized signal is not exactly the same as the original analog signal. There is a fair degree of quantization error here. However; as the number of quantization levels is increased the quantization error is reduced and the quantized signal gets closer and closer to the original signal

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# PCM of Speech Signals (very-important)

- Most of the significant spectral components of speech signals are contained in the range 300-3400 Hz
- Nyquist Rate =  $2 \times 3400 = 6.8$  kHz
- Practical Sampling Rate  $f_s = 8$  kHz (WHY..???)
- Number of quantization levels = 256

Number of Bits/Sample  $n = 8 (\log_2 256)$

Data Rate =  $nf_s = 8 \times 8000 = 64$  kbps

# PCM of Speech Signals (very-important)

- **Bandwidth Requirement**
- Communication theory tells us that *we can transmit error free at most two pieces of information per second per hertz bandwidth.*
- Therefore the minimum required bandwidth for transmission of a PCM speech signal  $BW_{\min} = 64/2 = 32 \text{ kHz}$